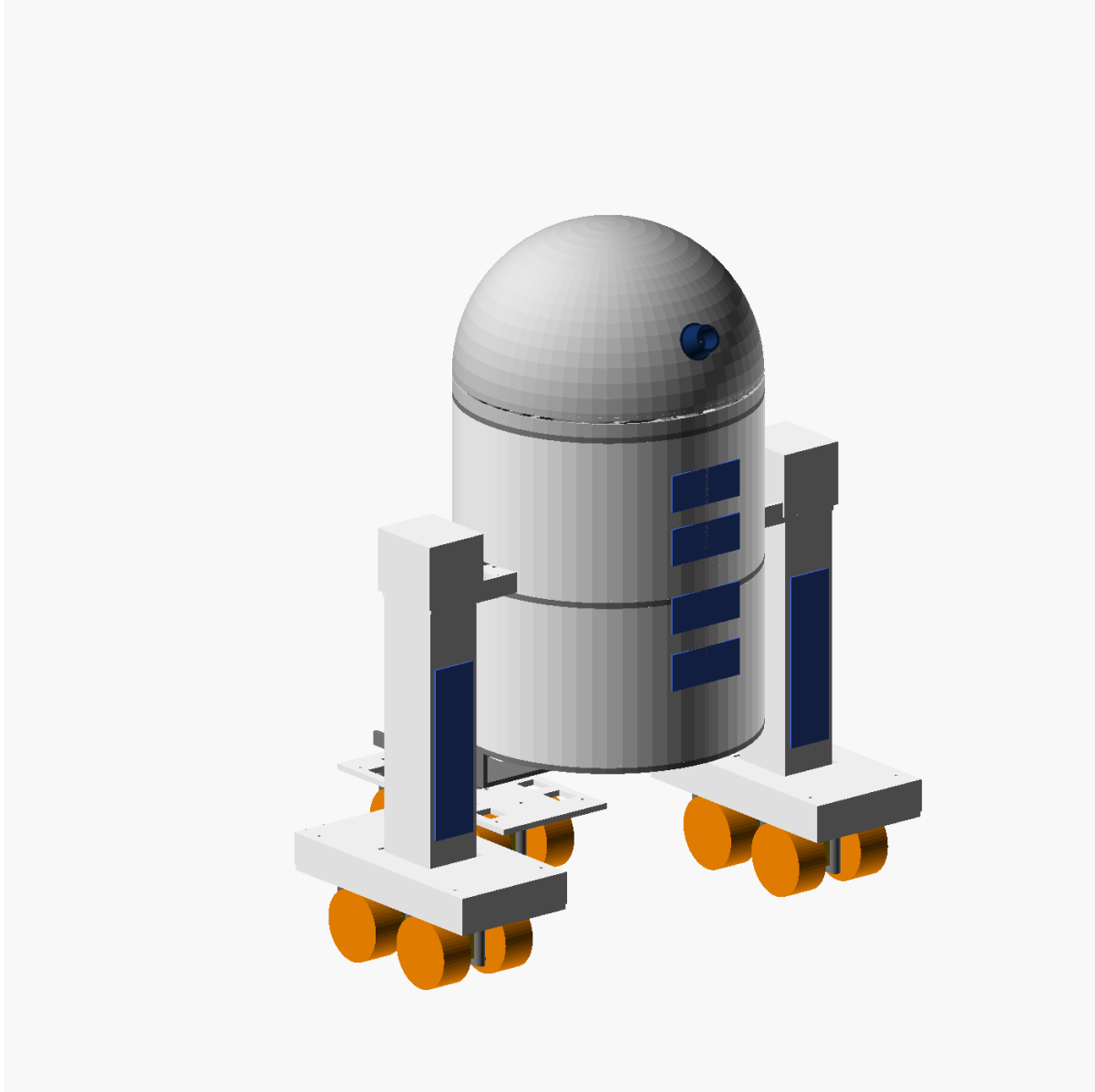


R2-24

Design and assembly manual

24 inch body • 6 motors • 12 wheels • powered rear steering • rotating head



September 11, 2026 | Revision A | H2D printable fabrication package

Digital checks passed. Physical fit, motor load, temperature, stopping distance and runtime still require the commissioning tests.
This PDF is not proof of a built or tested robot.

Package and reading order

39 distinct STL files; 148 printed pieces including fit parts and shim options; 16 original MP3 clips; 154 point-to-point wire connections.

Budget estimate: USD 570.93, including the listed hardware, consumables and four 1 kg filament spools. This mixes researched component prices with explicit allowances. It excludes shipping, tax, tools and printer. Unused fasteners and filament are expected.

Start with the mechanical and electrical chapters. Complete the 50 assembly steps in order. Stop at the loaded-chassis gate before printing all cosmetic skins. Use the final BOM, print manifest and circuit sheets during construction.

Contents

Mechanical design and print guide

Electrical design and wiring

Assembly and commissioning

Original robot voice collection

Digital verification and physical limits

Electrical BOM, hardware BOM and print manifest

Full wiring schedule

Circuit sheets

Mechanical design and print guide

Design envelope

Overall nominal height is 609.6 mm / 24 in, measured from the wheel contact plane to the dome crown. The body is 260 mm diameter. The side foot plates span 490 mm across X; covers increase this to 492 mm. The straight rear foot extends to Y=-290 mm and the body front reaches Y=+132 mm with trim. Allow about 500 x 430 mm floor space, plus turning clearance. Each exported part is smaller than 300 mm on every axis. The dome prints as one 260 mm diameter hemisphere; body and leg structures print in smaller sections.

Coordinates in CAD: X right, Y forward, Z up. Floor is Z=0. Main foot centers are X=-190/+190, Y=0. Rear steering center is X=0, Y=-180. Wheel axles are at local Y=-45/+45 and Z=31.5. All wheel axes start parallel to X. The nominal wheel centers are local X=-33/+33; actual hub insertion depth must be checked using the purchased wheels. Do not force a wheel against the yellow gearbox. Motor bodies face the same way within each foot.

Load and drive limits

Use only a dry, level, hard indoor floor. The design uses low-cost plastic gearboxes, not load-rated wheelchair drive units. Two wheels on one motor share its torque. A 63 mm wheel and the specified 0.8 kg.cm stall torque give about 2.49 N per motor at 6 V stall. Six motors would give 14.94 N at that unusable operating point. This build runs near 5 V with a 1 A driver limit, so available force is lower. There is no rated continuous torque curve or shaft radial-load rating in the product data.

Target a completed mass no greater than 4.5 kg. This is a commissioning limit to test, not a manufacturer-approved motor load. Read the actual sliced mass, then add the 0.7 kg battery, 0.456 kg wheels, 0.184 kg motors, bearings, metal hardware, wiring and boards. The geometry-only solid plastic bound in `cad/validation.json` is conservative for infilled parts but excludes support material. Do not add payload, thick filler or ballast.

At 4.5 kg, an assumed rolling resistance coefficient of 0.03 gives about

1.32 N straight-line resistance. Turning also scrubs the tandem wheels. That coefficient is only an engineering assumption. First test the rolling frame with securely attached temporary ballast up to the intended final mass. Require steady motion without stalls and no continuous current above

0.5 A per motor during gentle level travel. Stop a stuck motor promptly. If the frame cannot pass, reduce mass or revise the transmission before printing the full shell. Digital geometry cannot certify this test.

The rear foot is steered up to 25 degrees each way. The side drive ratio and rear speed follow the same circular path. Firmware does not request a zero-radius pivot. The drive ceiling is 110/255 duty and the phone starts at 35 percent of that ceiling. No encoder is fitted, so commanded speed and angle are not measured speed or angle. Calibrate steering with a protractor and observe actual path tracking.

Purchased fits and adjustable mounts

Motor 3777 uses the manufacturer drawing in the research file. Its gearbox thickness is about 18.6 mm and axle-tip span about 36.6 mm. The printed cradle supports the motor case with thin foam and two cable ties. It does not depend on uncertain case screw positions. The yellow case stays above the 3 mm floor. Set the axle center at 31.5 mm with the wheels touching a flat reference surface before tightening the ties. The cradle floor is at Z=16 and its deck support pillars end at Z=65. Keep all ties off both axles.

Bearings are 608-2RS, 8 x 22 x 7 mm. Tower pockets are 22.2 mm diameter. Print the fit coupon and one tower first. Seat bearings with hand pressure; ream uniformly if necessary. Do not drive a tight bearing into a printed tower with a hammer. Bearing shims are 8 mm ID, 12 mm OD, 1 mm thick so they contact only the inner race. Ordinary large M8 washers can bind the seals.

Both gear sets use module 1.5, 20 degree pressure angle, 20/40 teeth, 6 mm face width, and nominal 45 mm center distance. MCAD supplies an involute tooth profile. The servo mounting slots permit mesh adjustment. The 20 tooth pinion attaches to the supplied servo horn with M2 hardware; no printed spline is used. The 40 tooth hub has an M8 nut pocket to transmit torque to

the spindle. Use two jam nuts to lock the shaft after preload.

The servo mount drops its floor 8 mm below the deck. Set the horn/pinion height with the included 1, 2 and 4 mm shims. Keep the two gear faces aligned within 0.5 mm. The head pinion base is Z=461; rear pinion base Z=151. Servo case and horn revisions vary; check the actual height before final wiring. The rear bracket and head deck have clearance openings for the dropped cradle. Use 1 mm side foam to prevent movement, then tighten two ties.

Height stack and mating surfaces

The side foot deck is Z=65..70. Two 125 mm leg segments stack from Z=70 to

320. The shoulder bridge is Z=320..325, with stiffening ribs up to 338. Each pair of leg rods passes through the foot, both segments and the outer bridge holes. Inner bridge holes at X=+/-105, Y=+/-20 attach to the middle body frame. Decorative shoulder caps sit above the bridge and do not carry body weight. Foot covers clear the legs through their central openings.

Body frame lower faces are Z=170,315,460; each frame is 5 mm thick. Battery adapter is Z=175..178. Its 137 mm posts reach the middle frame at

315. The second adapter is Z=320..323. Four 97 mm posts reach the head deck at 420. Head deck is Z=420..424. Four 36 mm posts reach the upper frame at

460. The body rods at X=+/-75,Y=+/-75 pass through this complete stack. The utility deck sits on four 6 mm spacers at Z=329..332.

Lower cosmetic quarters span Z=175..315, and upper quarters Z=320..460. The upper quarters have shoulder slots on the X sides. Mirror X/Y to make all four upper pieces; do not rotate all four around Z. Two upper-bottom mount holes near the shoulder slots per side are intentionally absent. Use the eight surviving holes around that lower rim. Other shell rims have twelve holes. Shell flanges sit on frame faces, not inside them.

Rear attachment plate is Z=165..170, under the bottom frame. It picks up the two rear body rods. The bracket is centered at (0,-180,110); its upper flange ends at Z=165 and bolts under that attachment plate. Bearing tower starts at Z=114 and ends at 140; cap ends at 143. A 38 mm spindle sleeve starts on the rear foot boss at Z=75 and ends at 113. Add one 1 mm inner race shim before the lower bearing. Bearing-to-bearing sleeve is 12 mm. The upper bearing ends at 140, followed by one 1 mm inner shim and the hub at 141..160. Its teeth occupy Z=151..157. Adjust jam nuts for free rotation without axial play. The stationary cap clears the 12 mm rotating hub neck.

Head bearing tower is Z=424..450. The same inner race stack and upper shim put its hub at 451..470. Head plate sits at 470..473. Four 6.6 mm spacers at radius 119 mm carry the dome bottom at 479.6. The fixed neck ring is Z=465..477; it has 2.6 mm clearance below the dome. Head plate tips stay inside radius 124 mm. No light, speaker or wire is mounted on the spinning dome. Keep loose wiring below Z=420 and away from both gear pairs.

Print order and settings

1. Obtain the actual motors, wheels, servos and bearings. Print coupon, one motor_cradle, one bearing_tower, bearing_cap, race_spacer, servo_mount, servo_pinion, gear_hub, and all shim thicknesses first.
2. Check M3 clearance, M4 rod clearance, M8 shaft clearance, the 22.2 mm bearing pocket and the actual dual-wheel fit. Never scale an entire part to correct a hole; edit the relevant dimension and re-export that part.
3. Print structural PETG at 0.20 mm layers, four wall lines, five top/bottom layers and 20 percent gyroid as the starting settings. Load-bearing flanges, towers and gear teeth must contain continuous perimeters. Use dry filament. Check that the thin leg skins are fully filled by walls.
4. Print cosmetic PLA at 0.20 mm layers with a 0.4 mm nozzle and two wall lines. Body skins are 0.8 mm. Do not use vase mode. Let the designed flanges and bosses determine local thickness. Keep primer very light.
5. Use the exported orientation. Flat plates and ring sectors sit flat. Tall posts and leg segments stand upright. The dome opens downward. Support the dome roof, servo cradle flanges, gear-hub overhang and rear bracket upper shelf where Bambu Studio identifies unsupported surfaces. Remove support from bearing pockets and gear teeth before testing fit.

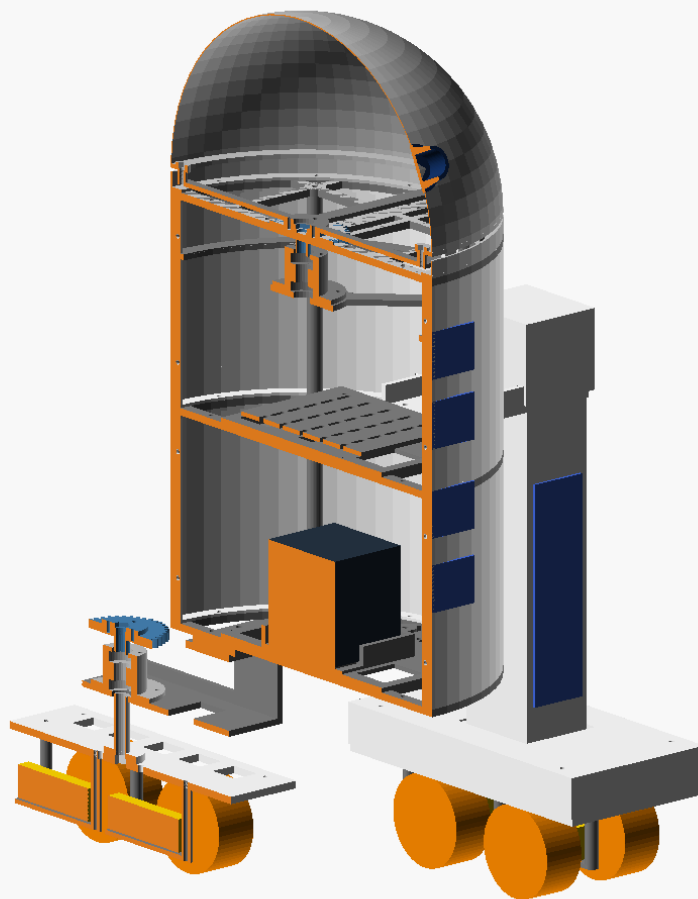
6. Frame quarters require four orientations by X/Y reflection. On each ring the small seam holes remain 5 mm from the seam. The upper shell needs the same reflected layout to keep shoulder slots on the sides. Mirror one shoulder bridge across X for the left side.
7. Print all quantities from `bom/printed-parts.csv`, including small spacers and the two side foot covers. Rear foot remains open for gear service; keep fingers away from its steering mechanism during operation.
8. Leave at least 5 mm around each part for brims. The H2D has a nominal 300 x 320 x 325 mm dual-nozzle volume; verify excluded regions in Bambu Studio. Do not send jobs to the printer until the first-fit checks pass.
9. Slice the full quantity list and record mass before printing all skins. Add real purchased-part mass and keep the completed robot below the 4.5 kg commissioning limit. The full plate layout was not physically printed or validated during this design run.

Finish and maintenance

Paint the body white, dome silver and trim blue. Mask eight blue dome panels separated by about 10 mm silver strips, plus a 12 mm blue band near the dome base. Paint the eye recess black. Use `detail_panel.stl` as a stencil for the body markings, or glue it with small conforming foam pads; its flat back needs about 5 mm edge compensation on the cylindrical shell. Paint the leg strips directly. These cosmetic parts are not structural.

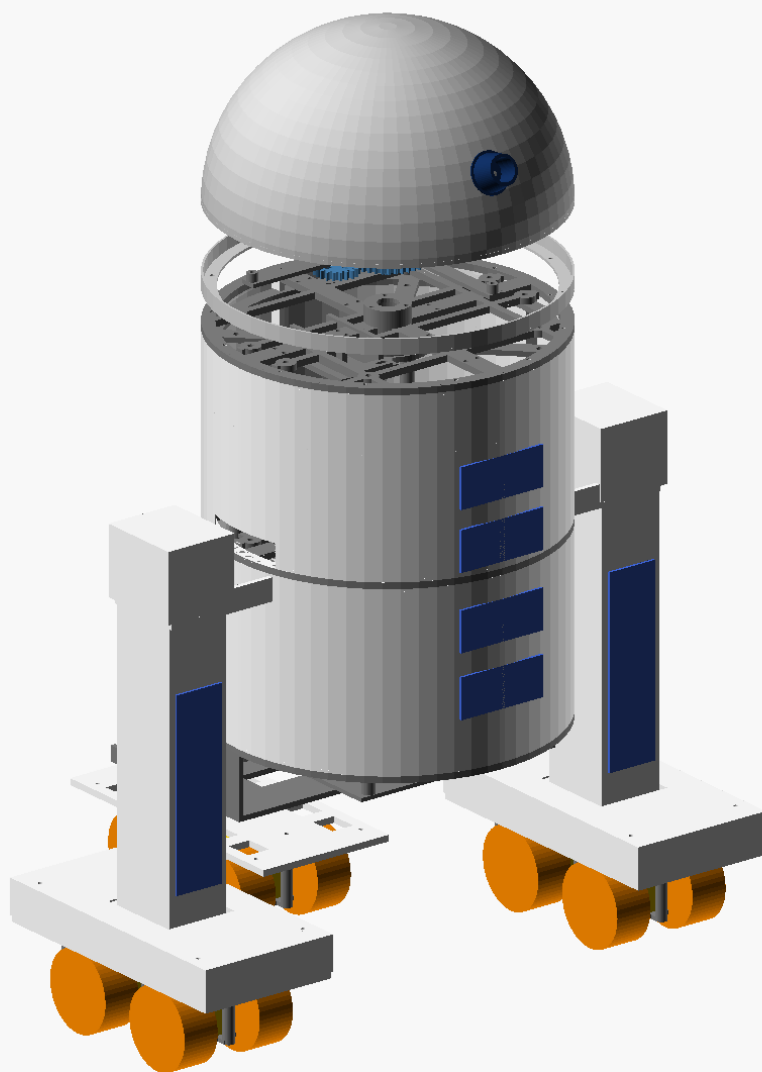
For service, isolate both switches and unplug the battery. Remove one rear upper shell quarter to reach the electronics. The dome lifts after its four retaining screws are removed. Recheck rod nuts, gear mesh, wheel press fits, cable ties and bearing play after the first hour, then before each session. Replace worn gearbox units rather than forcing a slipping wheel onto a cracked shaft. Do not lubricate belts or wheel treads; there are no drive belts in this design.

Internal frame and bearing stack



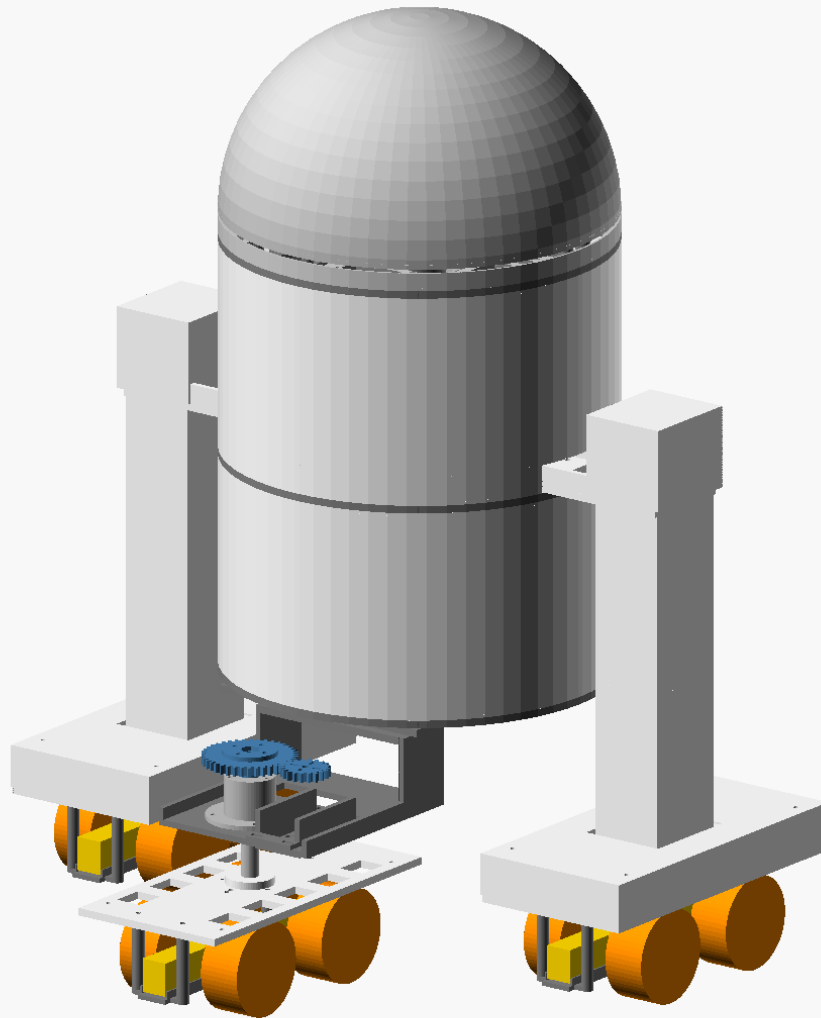
Section of the CAD: battery low in the body, rear steering under the frame, and a separate head bearing above the electronics.

Assembly groups



Groups are separated vertically for visibility. Use the height stack for actual positions.

Rear foot and service access



The rear foot stays open for gear and harness service. The switch faceplate is fitted to the chosen removable upper shell quarter during assembly.

Electrical design and wiring

Architecture

B1 is a protected Bioenno BLF-1206A 12 V, 6 Ah LiFePO4 pack with its matched

14.6 V charger. Adafruit supplies the six required TT motors, wheels, HUZAZH32 controller, three DRV8833 boards, four 5 V UBECs, logic regulator, servos and audio hardware. The battery, DC-rated switches, fuses, bearings and fasteners come from other suppliers where the required rating or mechanical part is not provided by the selected Adafruit range.

The circuit sheets are point-to-point diagrams with named nets. Each horizontal line is one electrical connection. Repeated net names join across sheets. `electronics/wiring.csv` is the full connection schedule, including all return wires. Do not interpret crossing text as a junction. Use the power overview for architecture and the detailed sheets for pins.

Six motor bridge channels are used. D1 drives the left pair, D2 the right pair and D3 the rear pair. Both channels on one driver receive the same direction signals, but each motor connects to its own output pair. Never connect two motors in parallel to one output pair. Retain the factory

0.2 ohm sense resistors, which set a nominal 1 A current limit per motor. Do not solder the current-limit bypass jumpers.

Power path

B1 positive passes through F1, a 7.5 A fuse within 100 mm of the mating connector, then S1 MAIN. MAIN powers the logic branch through a 1 A fuse. S2 RUN switches the input to four separate 2 A fused UBEC branches. P1, P2 and P3 power left, right and rear drivers. P4 powers both servos and the signal buffer. All grounds join the battery-negative distribution point. The positive 5 V outputs remain separate; never parallel them.

The three motor rails each allow 2 A maximum under the driver limits, which is below the UBEC's nominal 3 A capacity. Allocate up to 2 A for the two servo actuators as a commissioning budget; their simultaneous loaded current must be measured. Use the separate 1.2 A logic converter for ESP32 and low-volume audio. Avoid feeding the motors from the Feather USB pin, 3V pin, a solderless breadboard, or a small phone power bank.

At the conservative load budget of 30 W motors + 10 W servos + 5 W logic, 85 percent assumed conversion efficiency and 11.2 V battery voltage, input current is about 4.73 A. The pack is rated 12 A continuous, and the main fuse is 7.5 A. These figures size the supply; they do not permit stalled motors or servos. The 72 Wh pack at an assumed 15-25 W average load gives roughly 2.0-3.3 hours using 70 percent of nominal energy. This is an estimate only. Measure real runtime and temperature on the finished robot.

The UBEC input specification is 6-16 V; the fully charged pack is 14.6 V. Use the specified input capacitors, short power leads and no higher-voltage pack. Check the actual UBEC label and output polarity before connection. Output tolerance is about 5 percent; measure 4.75-5.25 V on each unloaded rail. Stop if a motor rail exceeds 6 V. The Feather logic rail must remain within its 5 V USB supply requirement; do not apply raw pack voltage.

Pin map

U1 is the original HUZAZH32 3405, using ESP32 GPIO numbers:

- GPIO14 / 32: left forward / reverse to D1 AIN1+BIN1 / AIN2+BIN2.
- GPIO15 / 33: right forward / reverse to D2 AIN1+BIN1 / AIN2+BIN2.
- GPIO27 / 12: rear forward / reverse to D3 AIN1+BIN1 / AIN2+BIN2.
- GPIO13: all three SLP inputs; external 10k pull-down. Low disables drive.
- GPIO36 / A4: all three open-drain FLT outputs; 10k pull-up to 3.3 V.
- GPIO25 / A1: rear servo pulse through AHCT125 channel 1.

- GPIO26 / A0: head servo pulse through AHCT125 channel 2.
- GPIO39 / A3: RUN power sense from the servo 5 V rail through 10k/15k.
- GPIO34 / A2: battery ADC through 100k/22k from switched MAIN power.
- GPIO18 / 19 / 23: amplifier BCLK / LRC / DIN, respectively.

Each of the six motor input signals has a 10k pull-down. GPIO12 is an ESP32 boot-configuration pin; do not attach any pull-up to it. GPIO34,36,39 are input-only, which is appropriate here. All logic is 3.3 V. No 5 V or raw battery line connects directly to a GPIO. The Feather BAT/JST connector remains empty; it is for a single-cell battery, not this pack.

Servo and sense circuits

U7 is a 74AHCT125 DIP14, powered from P4 5V_SERVO on pin14, ground pin7. Pin1 and pin4 output-enables go to ground. GPIO25 enters pin2 and exits pin3 through 220 ohms to SV1 signal. GPIO26 enters pin5 and exits pin6 through 220 ohms to SV2 signal. Pin10 and pin13 are tied to 5V_SERVO to disable unused channels. Pin9 and pin12 go to ground; outputs pin8 and pin11 remain unconnected. Put a 100 nF capacitor at pin14/pin7 and a 10k pull-down on each input signal. Check the notch before inserting the IC. Use the exact AHCT type with inputs rated independently of VCC; do not substitute HC logic or a board with undocumented input clamp circuits.

Servo red wires go to 5V_SERVO, brown/black to ground. Signal wire colors can vary, so follow the servo label. The board sends neutral head pulses while disarmed whenever RUN power exists. The rear steering holds its last angle. A physical RUN-off removes motor and servo power independently of the phone and firmware. It is a manual disconnect, not a certified safety system or a brake. Test actual stopping distance at low speed.

For battery sensing, connect MAIN through 100k to GPIO34, then 22k to ground. Put 100 nF from that junction to ground. At 14.6 V, the ADC sees

2.633 V nominal. Code scales by 122/22, cuts drive below 11.2 V for one second, and requires at least 12.0 V to re-arm. Calibrate against a multimeter using PACK_CORRECTION. The pack BMS remains the final battery protection and must not be bypassed.

For RUN sensing, connect P4 output through 10k to GPIO39, then 15k to ground, with 100 nF to ground. This makes about 3 V from a 5 V rail. No movement can be armed until RUN sense, battery and driver-fault status are healthy. A fault or lost heartbeat clears the movement lease; restored power or Wi-Fi does not automatically re-arm it.

Audio and USB

U8 MAX98357A VIN goes to 5V_LOGIC. GND goes to the common return. The speaker connects only across U8's two speaker terminals. Neither terminal goes to ground. Leave SD at the board default. Connect GAIN through 100k to 5V_LOGIC for low gain, and retain firmware gain 0.22 for initial tests. Use short I2S wires, ideally below 150 mm, and keep them away from motors.

A removable link J_USB connects U6's 5 V output to the Feather USB pin. Remove that link and unplug the robot battery before connecting a computer USB cable. Otherwise the two 5 V sources can feed each other. Flash and read the initial private Wi-Fi password with the controller disconnected from the robot power harness. Then remove USB, restore J_USB and reconnect the battery. Label the link clearly. Do not rely on remembering which source is connected.

Harness and charging

Use 18 AWG stranded copper for main distribution, 22 AWG for motor/servo branches and 26 AWG for signals. Fit ferrules to stranded wires in screw terminals. Twist each motor pair and put a 100 nF capacitor directly across the motor tabs. Add the listed electrolytics near their rails, observing polarity. The supplied motor leads are short and thin; strain-relieve the joint when extending them. Keep the six motors on separately labeled two-pin plugs M1 through M6.

Leave a restrained service loop at the rear steering pivot for only plus/minus 25 degrees, with enough slack for plus/minus 30. Route it clear of the gears and spindle. No conductor crosses the head's rotating joint. Secure fixed wires to the body rods and keep them away from wheel treads. Never secure wires to a rotating shaft. No bare conductor may touch a rod.

Turn MAIN and RUN off, unplug and remove the battery before charging. Use the factory charge connector and matched Bioenno LiFePO4 charger. Do not charge through the Feather, USB link, motor wiring or a LiPo charger. Do not open the battery pack or modify its BMS. Strap the battery in the tray with padding and two independent hook-and-loop straps after charging.

Sources

Primary references: [DRV8833 pin guide](#), [HUZZAH32 pin guide](#), [UBEC rating](#), [logic regulator](#), [MAX98357A](#), [Bioenno pack](#), [TI SN74AHCT125 pin functions and ratings](#). Exact purchased parts and price allowances are in the BOM. Prices and availability were researched on 2026-09-11 and are not a purchase quote.

Assembly and commissioning

Tools and first checks

Use a Bambu H2D, 0.4 mm nozzle, dry PETG and PLA, calipers, a flat reference surface, small square, protractor, M2/M3/M4 drivers, 7 mm and 13 mm spanners, side cutters, wire stripper, soldering iron, heatshrink, multimeter and a current-limited bench supply. A small luggage scale helps measure pulling force. A thermocouple or infrared thermometer helps check motor/regulator temperature. Eye protection is needed when drilling or cutting rods.

Read the mechanical height stack and electrical chapter before assembling. Inspect purchased items against the BOM. Print and test the fit parts first. Keep the battery disconnected until the electrical checks reach the specified connection step. Do not start with the complete shell installed; the first chassis test determines whether the chosen motors can move the actual weight on your floor.

Stage 1 - feet and motors

1. Label foot plates L, R and B. Mark their forward Y direction. Both motors on each plate use the same orientation, with the motor cans toward negative Y and shafts at local Y=-45 and +45.
2. Smooth the six cradle floors and remove support from the slots. Put thin foam under each motor. Feed two 3 mm cable ties through the slots at local Y=-40 and -15. Leave them loose.
3. Fit one wheel on each motor shaft. Support the opposite axle while pressing by hand. Do not hammer or grip the gearbox with pliers. Keep equal insertion depth on the two sides and leave clearance to the case.
4. Place each cradle at Z=16 with wheels on a flat reference plane. Adjust the thin padding so both axle centers are about 31.5 mm above that plane. Rotate both wheels by hand; they must clear rails, ties and the case.
5. Fasten each cradle under its foot deck with three M3x60 bolts, washers and nuts. Pillar holes are at X=-18/+18,Y=-50 and X=0,Y=+20 relative to each shaft. Pillar tops meet the deck underside at Z=65. Do not crush the pillars; tighten until the washers stop moving.
6. Tighten the case ties only enough to prevent motion. Check shaft and wheel clearance again. Motor M1/M2 belong to L, M3/M4 to R, M5/M6 to B. Solder each suppression capacitor across its motor tabs and label leads.
7. Fit the two side foot covers. Their four integral posts land on the deck. Use four M3x35 bolts per cover at X=+/-46,Y=+/-90 with washers and nuts. Leave the center leg openings clear. The rear foot has no cover so the steering spindle and harness remain accessible.

Stage 2 - body frame and side legs

8. Build three body rings from four frame_quarter pieces each. Mirror pieces across X/Y as shown by the assembly view. The paired seam holes are 10 mm apart across each seam at radii 40 and 95. Fit two splice plates per seam, each with two M3x16 bolts, washers and nuts. Use eight splices per ring, twenty-four total. Assemble on a flat surface.
9. Put the bottom ring at Z=170. Place rear_attach under its rear edge at Z=165..170. Its outer holes join the rear body rods at X=+/-75,Y=-75. Put the lower deck_adapter on the ring at Z=175..178.
10. Cut four M4 body rods to 315 mm and deburr both ends. Fit nuts and washers under the bottom ring, including the rear attachment plate on the rear two rods. Add four 137 mm body_post sleeves above the lower adapter. Add the middle ring at Z=315..320 and the second adapter at Z=320..323. Snug the lower nuts; do not preload the frame heavily.
11. Add four 97 mm head_support sleeves, the head_deck at Z=420..424, four 36 mm top_post sleeves and the upper body ring at Z=460..465. Use the diagonal rod holes in the head deck, not its servo slots. Fit washers and double nuts above the top ring. Check all three rings are parallel, centered and square to the rods before locking the nuts.
12. Cut four M4 leg rods to 280 mm. Two go through each side foot at X=0,Y=+/-20. Stack two leg_segment parts over each pair from Z=70 to 320. The 1.2 mm skins are covers around the rod structure; do not clamp or lift the robot by squeezing them.

13. Put each shoulder bridge on its leg at Z=320. Mirror the left bridge. Fit the outer rod nuts and washers. Its inner two holes at X=+/-105, Y=+/-20 bolt to the matching middle ring holes with M4x20 hardware. Keep both feet parallel. Add the decorative shoulder caps only after the structure passes the rolling test; use small removable adhesive pads rather than glue on the load-bearing joints.

14. Bolt the battery tray to the lower adapter with four M3x12 bolts at X=+/-45, Y=+/-30. Leave the battery out. Add two straps through the tray slots and verify the discharge and charge leads have a free exit.

Stage 3 - rear steering and bearings

15. Fit the rear_bracket under rear_attach using the four M4x20 joints at X=+/-35, global Y=-103/-82. The rear bracket origin is (0,-180,110). Its 5 mm top flange ends at Z=165. Check the bracket is square to the body and the spindle center is exactly 180 mm behind the main feet.

16. Press two 608 bearings into a bearing_tower, one from each end. Insert the 12 mm race_spacer between their inner races. Mount the tower on the rear bracket at Z=114 using four M3x16 bolts through radius-24 holes. Add the cap with three M3x8 thread-forming screws in the 2.6 mm pilot holes. Do not use machine screws in these pilots.

17. Insert the M8x120 rear spindle bolt from below the rear foot, seating its head in the hex recess. Fit the 38 mm spindle_sleeve on top of the foot's 10 mm central boss, then one 1 mm inner-race shim. Feed the shaft through the lower bearing, internal spacer and upper bearing.

18. Add a 1 mm inner-race shim above the upper bearing, then gear_hub. Seat one thin M8 nut in its hex pocket and add a second jam nut. Snug the stack until axial play disappears while the foot turns freely. Hold the inner nut and lock the outer one. Do not tighten against the bearing outer rings or seals. Trim excess bolt projection if needed, keeping at least two full threads beyond the final nut.

19. Power the steering servo alone with a suitable 5 V bench supply and a 1500 microsecond test pulse, or use the firmware with all drive motor plugs disconnected. Center the servo before attaching its horn. Do not move the servo by forcing the horn against the gearbox.

20. Bolt its supplied horn to a servo_pinion with M2 bolts/nuts through the radial slots. Use the supplied spline center screw. Place the servo in its dropped cradle and set the pinion base to Z=151 with the 1/2/4 mm shims. Secure two ties and foam pads without crushing it.

21. Bolt the cradle into the rear bracket slots. Align the rear foot straight ahead before meshing the gears. Slide the cradle to obtain free mesh with slight backlash and no tooth bottoming. Nominal center distance is 45 mm. Tighten the four mounting bolts only after checking the full intended steering range.

22. With motors still unplugged, command rear steering slowly in each direction. At phone +100/-100 the actual foot must reach +25/-25 degrees, within about two degrees. Adjust STEER_CENTER_US and STEER_US_PER_DEGREE in config.h to the measured horn/servo response. Gear inversion is already accounted for by the negative pulse slope. If the foot binds or stalls, stop and fix the geometry before travel.

23. Route rear motor wires through a restrained service loop that permits plus/minus 30 degrees without touching the gears. The loop must not tighten at the command limits. Do not route wires through the spindle.

Stage 4 - rotating head

24. Assemble the second tower and bearing spacer as in step 16. Mount it on head_deck at Z=424. Its cap ends at Z=453. Keep cap screw ends away from the bearing shields.

25. Insert the M8x65 bolt from below the lower bearing, with a 1 mm inner-race shim under its head. Add the upper shim and gear hub. Fit the captured nut and jam nut. The hub starts at Z=451 and ends at 470. It must rotate freely without rocking; adjust preload by hand.

26. Fit the FS5103R continuous servo in its cradle with its supplied horn and pinion. Set the pinion base to Z=461 with shims and align the teeth. Adjust the servo's neutral trim so 1500 microseconds produces no movement. Record any remaining neutral

correction in config.h.

27. Attach head_plate to the hub's four radius-12 holes with M3x18 bolts, washers and nuts. Its central opening clears the spindle nuts. Check that no screw head contacts the cap during a full rotation.

28. Fix neck above the upper frame at Z=465 with twelve M3x16 joints. Fit the dome's four M3 captured nuts. Put four 6.6 mm spacers between head plate and dome, centered at radius 119. Use four M3x16 bolts from below into the dome nuts. The dome starts at Z=479.6.

29. Turn the head manually through 360 degrees with power off. Require at least 2 mm clearance around the fixed neck and no contact between rods, screw ends and rotating parts. Then run the head at low speed in both directions. No wire belongs inside the dome. The eye and blue markings are decorative and unpowered.

Stage 5 - boards, wiring and inspection

30. Mount the utility deck on four 6 mm spacers above the middle adapter. Use four M3x20 through bolts. Place the HUZZAH32 near the service opening, with its USB socket accessible. Use M2.5 bolts in board holes where provided, and foam plus cable ties on slots where a board has no usable hole. No exposed solder joint touches a metal rod or deck.

31. Mount the three driver boards along the utility deck edge, keeping each motor plug labeled. Put the buffer/prototype board and amplifier away from driver output wires. Keep I2S leads short. Mount the speaker in its printed plate using a soft rim pad and two ties through the side slots; do not press on the cone. Fasten or tie the speaker plate to unused utility-deck slots, cone up, clear of the head mechanism.

32. Place the four UBECs on the lower adapter in pairs near Y=±58, outside the battery tray. Restrain them with ties through the adapter openings. Keep exposed cases apart and allow air around their surfaces. The logic buck and fuse holders can use the remaining lower deck edge.

33. Choose the rear upper shell quarter as the removable service panel. Use switch_plate as the drilling template: two 12.5 mm openings centered 30 mm apart, and four M3 attachment holes on a 60 x 32 mm rectangle. Place the panel center near global Z=365 and away from a seam or rod. Drill the shell only after checking the switch-body and wire clearances inside. Use the switch plate as a flat backing face with small foam pads at the curved-shell edges. Label MAIN and RUN.

34. Wire every connection in wiring.csv, marking rows complete. Work through power first, driver signals next, buffer/servos, sense and audio last. Check resistor values with the meter before installation. Keep battery positive disconnected and fuses removed during soldering.

35. Inspect every joint, ferrule and capacitor polarity. Check that none of the four 5 V UBEC outputs is connected to another positive output. Verify all grounds connect. Verify speaker leads are isolated from ground. Inspect the AHCT notch and all fourteen pins against the map.

36. With boards and motors unplugged, test power converters using a current-limited 12 V bench supply. Measure each 5 V rail, MAIN and RUN switch behavior and both divider outputs. Pack-sense ratio is 22/122; RUN-sense ratio is 15/25. Disconnect power before attaching boards.

37. Restore the boards, keep motors and servos unplugged, and test again at a low current limit. Confirm 3.3 V on the Feather and a high FLT input. Correct unexpected current or heating before moving on.

Stage 6 - firmware and first drive

38. With robot battery disconnected and J_USB removed, connect the HUZZAH32 to the computer. Run `pio run -d firmware -t upload`, followed by `pio run -d firmware -t uploadfs`. Use `--upload-port COMx` for the actual controller port if automatic detection is ambiguous. Do not upload to another device connected to the computer.

39. Open `pio device monitor -b 115200`, reset the board and privately save its generated Wi-Fi password. Join network R2-24 from the phone and open <http://192.168.4.1>. This is a local access point; it needs no router or Internet account. The initial screen must be disarmed.

40. Disconnect USB, restore J_USB and reconnect robot power with RUN off. Strap the battery down with both straps. Compare displayed voltage to a meter, then calibrate PACK_CORRECTION if needed. Arming must fail while RUN is off. Turn RUN on only with all wheels lifted and clear.
41. Connect one motor pair at a time. Arm, hold forward at minimum speed and check that both wheels on each motor move forward at the ground. Reverse motor plug polarity if one motor differs within its pair. Change the group's DIRECTION in config.h if the complete pair is reversed. Disconnect power before changing wiring.
42. Test reverse, release-to-stop, STOP, loss of phone Wi-Fi, page hide, browser close and RUN-off. Motion commands expire within 500 ms of the last accepted heartbeat. No old command or power restoration may restart motion without a new Arm press. Check actual wheel stopping time as well as command timeout. Physical RUN-off must always work.
43. Reconnect servos and verify the steering and head calibration from stages 3-4. Head motion stops at release and at STOP. The rear foot holds its current angle after STOP; it does not snap to center. Use Center steering while armed and stopped to align it for handling.
44. Play all sixteen sound clips at low volume. Confirm no controller resets when audio starts or motors accelerate. A reset means power integrity needs repair, not a reason to disable the fault checks.
45. Place the bare chassis on the intended floor, with a person beside the physical RUN switch. Test short straight travel, reverse and large arcs at low speed. The rear wheels must follow the curve without skidding sideways. Do not test stairs, carpet, slopes or thresholds.
46. Secure temporary ballast low in the tray area to reach the predicted completed mass, at most 4.5 kg. Measure current and temperatures during several gentle arcs and a ten-minute run. Require no stalls, no driver faults, no supply resets, no loose shafts and no continuous motor current above 0.5 A during gentle travel. Stop if cases exceed 50 C, wiring heats or gears grind. These are conservative build acceptance gates, not published component operating ratings.

Stage 7 - skins and final acceptance

47. Only after the loaded chassis passes, install the lower four shell quarters. Use M3x12 at the bottom rim and vertical seams. Use M3x16 where both shell flanges sandwich the middle frame, and through the upper shell/frame/neck stack. Install the four mirrored upper quarters around the shoulder slots. Use its eight remaining lower-rim holes; the other four middle-ring joints hold only the lower shell with M3x12. Use all twelve upper-rim holes. Keep the rear service quarter removable. Require full nut engagement and no screw tip touching a rotating part; shorten only excess tips after dry assembly.
48. Complete the light paint finish and attach the eye with a small epoxy fillet after fitting it to the dome curvature. Use the printed panel as a paint stencil or apply with compliant pads. Do not add thick resin coatings or filler. Do not glue service screws, bearings or gears.
49. Weigh the completed robot and measure height. Target 609.6 mm, always below 914.4 mm. Confirm the measured mass does not exceed the load used in the successful chassis test or the 4.5 kg commissioning limit.
50. Repeat the stop, disconnect, steering, sound and ten-minute temperature checks with all skins installed. Record current, temperatures, mass, stopping distance, steering limits and runtime in a build log. Mark the machine ready only after all tests pass. Keep it supervised.

Fault finding

If the controller will not arm, check RUN sense, pack voltage, the common ground and open-drain FLT wiring. If it resets under load, inspect power distribution and USB-source isolation. If a foot turns opposite the others, check motor polarity before changing the mixer. If the rear foot scrubs, check its mechanical neutral and servo slope. If the dome drifts at rest, trim the continuous servo neutral. If a bearing binds, remove oversized washers or reduce preload. If the robot cannot move the test mass, do not raise current limits to hide it; revise weight or gearing and repeat the chassis gate before closing the shell.

Original robot voice collection

The sixteen MP3 files are in `firmware/data/audio/`. They are original synthesized beeps, chirps and whistles. They are not recordings from a film, and they do not reproduce an actor's voice. Names describe intended moods.

`catalog.csv` provides names, durations and SHA-256 hashes. The phone sound menu plays every file through the onboard amplifier and speaker. The collection includes greetings, questions, agreement, concern, excitement, sleepy sounds, searching, surprise and a longer robot story.

Format: mono MP3, 22,050 Hz, 64 kbit/s. Rebuild with `python scripts/generate_audio.py`; NumPy and FFmpeg are required. The deterministic oscillator source and seeds are included. No web audio was downloaded. These generated audio files may be used, changed and shared with this robot project. No claim is made to the R2-D2 trademark.

After changing audio, run `pio run -d firmware -t buildfs`, then upload the filesystem to the disconnected controller as described in the manual. Firmware playback gain is limited to 0.22. Start with a low listening volume.

Digital verification and physical limits

The fabrication package passed digital checks. No printer, completed robot, motor load, battery runtime or physical wireless link was tested during this run. The staged commissioning tests remain necessary before operation.

CAD

`python scripts/export_cad.py` produced 39 different STL meshes and a manifest for 148 pieces, including first-fit parts and shim options. All meshes passed closed surface, consistent winding, positive volume, one connected solid and 300 mm per-axis checks. The geometry-only solid plastic bound is 2924.6 g. This is not a measured print mass or the total robot mass.

`python scripts/verify.py` also evaluated OpenSCAD intersections for the shoulder bridge/upper shell, head plate/neck, rear spindle sleeve/bracket, and a 20/40 tooth gear pair at five mesh phases. Each intersection was empty. These selected checks do not establish clearance for every purchased part revision, every fastener projection or a misrouted harness.

Bambu Studio also sliced `coupon.stl` with the installed H2D 0.4 mm profile,

0.20 mm Standard process and PLA Basic. It returned code 0, Success., and no warnings, and produced real H2D G-code inside a sliced 3MF. Its predictions were 12.36 g and 1436 seconds for that coupon. The test used isolated temporary settings and never connected to a printer. This one stock-PLA coupon slice does not validate every structural PETG profile or predict the full robot's mass. Evidence: `cad/h2d-slice-check.json`.

Firmware and sounds

`pio run -d firmware` succeeded for the original Adafruit HUZZAH32 board, using espressif32 6.5.0 and Arduino-ESP32 2.0.14. `pio run -d firmware -t buildfs` succeeded with the web UI and sixteen MP3s in a 2 MiB filesystem. Both binary images fit their declared partitions. They have not been uploaded to a physical controller as part of this design task.

`python scripts/verify.py` compiles the actual controller header with MinGW g++ using `-std=c++11 -Wall -Wextra -Werror`, and executes the result. It protects arming lockout, stale leases, replay rejection, command timeout, fault response, timer rollover, drive/steering bounds and reversal delay. The phone test runs the real JavaScript with a simulated page/network and checks release, cancel, focus loss, delayed arm replies and disconnects. Node syntax checking also passes.

All sixteen MP3s have matching SHA-256 hashes and fully decode in FFmpeg. They contain 42.1 seconds of original synthesized robot sounds. MP3 playback on the physical ESP32 and acoustic output still need the build tests.

Electrical consistency

The 154-row wiring schedule has six independent motor output channels. The verification script compares all paired direction pins and control constants with the firmware. It checks the pack and RUN divider upper bounds remain below 3.3 V at the stated voltages. These are static checks, not continuity measurements on a soldered harness. Source parts, ratings, wire sizes, fuses and mounting consumables are included in the BOM.

Physical acceptance still required

Measure purchased-part fit, wheel press-fit retention, actual steering angles, gear mesh, bearing preload, shell mass, completed mass, supply voltage/current, temperatures, stopping distance and runtime. Require the rolling chassis to pass with the intended final load before printing and fitting all skins. Stop if the motor load or thermal tests fail. Do not describe this package as a tested functioning robot until those tests pass.

Machine-readable digital check output is in `docs/verification.json`.

Electrical bill of materials

USD prices are per item. Listed prices were researched on September 11, 2026; allowances are estimates. Follow the exact rating and part notes.

Qty	Component	Unit USD	Fit / rating / source
6	M1-M6 TT motor Adafruit 3777	\$2.95 (listed)	3-6V; six separate current-limited channels Source
12	W1-W12 63mm TT wheel Adafruit 3766	\$1.50 (listed)	Listed out of stock; two wheels per motor Source
1	U1 HUZZAH32 original Feather Adafruit 3405	\$19.95 (listed)	Original ESP32; not ESP32-S3 or V2 Source
3	D1-D3 Dual DRV8833 driver Adafruit 3297	\$5.95 (listed)	Leave factory 1A current limits intact Source
4	P1-P4 5V3A UBEC Adafruit 1385	\$9.95 (listed)	Separate left right rear and servo rails; do not parallel Source
1	U6 5V1.2A logic converter Adafruit 4739	\$6.95 (listed)	Logic and audio only Source
1	SV1 SG-5010 standard positional servo Adafruit 155	\$11.95 (allowance)	Rear steering; use supplied horn and center screw Source
1	SV2 FS5103R continuous rotation servo Adafruit 154	\$11.95 (listed)	Head speed control; check stock and neutral trim Source
1	U7 74AHCT125 DIP14 Adafruit 1787	\$1.50 (allowance)	5V servo signal buffer; use exact AHCT type Source
1	U8 MAX98357A I2S amplifier Adafruit 3006	\$5.95 (listed)	Bridge output; neither speaker terminal is ground Source
1	SP1 3inch8ohm1W speaker Adafruit 1313	\$1.95 (listed)	77.8mm square envelope; foam and cable tie mounting Source
1	B1 Protected12V6AhLiFePO4 battery Bioenno BLF-1206A	\$79.99 (listed)	72Wh;12A continuous;0.7kg;112x66x71mm Source
1	CH1 Matched14.6V2A charger Bioenno BPC-1502DC	\$21.99 (allowance)	Remove battery for charging; use factory charge jack Source
2	S1-S2 20A30VDC SPST panel switch NKK S1A	\$12.00 (allowance)	MAIN and RUN;12.5mm panel holes; label OFF clearly Source
6	FH1-FH6 ATO inline fuse holder16AWG Littelfuse 0FHA0001ZXJ	\$4.00 (allowance)	Insulated holder; no exposed blades Source
1	F1 7.5A32V ATO fuse ATO7.5A	\$1.00 (allowance)	Main pack fuse
4	F2-F5 2A32V ATO fuse ATO2A	\$1.00 (allowance)	Each UBEC input separately fused
1	F6 1A32V ATO fuse ATO1A	\$1.00 (allowance)	Logic input fuse
1	PP30 Anderson mating plug set PP30	\$6.00 (allowance)	Match factory discharge plug; do not cut pack lead Source
2	PROTO Half-size solderable board Adafruit 1609	\$4.50 (allowance)	Buffer and low-current distribution; motors use heavy wire Source
1	SOCKET DIP14 socket 14pin0.3inch	\$0.75 (allowance)	For U7
11	R10K 10kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Six PWM;enable;fault;two servo inputs;power sense top
1	R15K 15kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Power sense bottom
2	R100K 100kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Pack sense top and amplifier gain
1	R22K 22kohm1percent0.25W resistor Metal film	\$0.10 (allowance)	Pack sense bottom
2	R220 220ohm0.25W resistor Metal film	\$0.10 (allowance)	Servo output series resistors
9	C100N 100nF25V ceramic capacitor X7R	\$0.15 (allowance)	Six motor suppression;buffer bypass;two sense filters
4	CI1-CI4 220uF25V electrolytic 105C low ESR	\$0.75 (allowance)	UBEC inputs; observe polarity

Qty	Component	Unit USD	Fit / rating / source
1	CI5 100uF25V electrolytic 105C low ESR	\$0.50 (allowance)	Logic converter input
4	CO1-CO4 470uF10V electrolytic 105C low ESR	\$0.75 (allowance)	Three driver rails and logic rail
1	CS 1000uF10V electrolytic 105C low ESR	\$1.00 (allowance)	Servo rail
4	WIRE18 18AWG silicone wire metre Red black	\$1.50 (allowance)	Main power distribution
6	WIRE22 22AWG silicone wire metre Assorted	\$1.00 (allowance)	Motor and servo branches
5	WIRE26 26AWG stranded wire metre Assorted	\$0.60 (allowance)	Signal wiring
1	CONNECTORS Connectors ferrules headers heatshrink and labels Allowance	\$22.00 (allowance)	Six removable motor plugs;two servo extensions;USB power link;10A terminal blocks
1	USB Micro USB data cable USB-A to micro-B	\$5.00 (allowance)	For original HUZZAH32 flashing

Mechanical hardware and consumables

Metal bolts, bearings and rods are purchased, not printed. STL mounts and adapters are in the print manifest. Quantities include stated spares and consumables.

Qty	Item	Specification	Use
90	M3x12 bolt	Steel machine screw	Shell seams and single-flange joints; count includes spares
18	M3x60 bolt	Steel machine screw	Three per motor cradle through deck
8	M3x35 bolt	Steel machine screw	Side foot covers
90	M3x16 bolt	Steel machine screw	Splices; double shell flanges; neck; bearings and speaker
16	M3x20 bolt	Steel machine screw	Servo cradle slots and switch faceplate
4	M3x18 bolt	Steel machine screw	Head plate through gear hub; clear stationary cap
4	M3x16 bolt	Steel machine screw	Dome spacers to captured dome nuts
6	M3x8 thread forming screw	For 2.6mm plastic pilot holes	Bearing caps; only these thread into plastic
210	M3 hex nut	DIN934 steel	Through joints; count includes spares
360	M3 flat washer	Steel 3.2mm ID	Each bolt head and nut on plastic
4	M4 threaded rod	4mm diameter x315mm steel	Body spine; cut and deburr
4	M4 threaded rod	4mm diameter x280mm steel	Two through each side leg
8	M4x20 bolt	Steel machine screw	Shoulder inner holes and rear attachment
40	M4 hex nut	Steel regular plus jam nuts	Body and leg rods; attachment
64	M4 washer	Steel 4.3mm ID	Spread load on plastic
1	M8x120 fully threaded bolt	Steel with 13mm hex head	Rear steering spindle; verify useful thread length
1	M8x65 fully threaded bolt	Steel with 13mm hex head	Head spindle
8	M8 thin jam nut	Steel 13mm AF	Spindle lock pairs; no adhesive on bearings
8	M8 bearing shim	8mm ID x12mm OD x1mm	Contact inner races only; not ordinary wide washers
4	608-2RS bearing	8x22x7mm	Two rear and two head
8	M2x8 bolt and nut set	Steel	Servo supplied horns to printed pinions; adjust to horn thickness
20	M2.5x16 bolt and nut set	Steel	Boards on printed insulating spacers
4	M3x20 bolt and nut set	Steel	Utility deck on 6mm spacers
2	Hook and loop strap	20x300mm	Battery retention through tray slots
40	Cable tie	3.0mm width x200mm	Motor servo and PCB retention; plus wiring restraint
1	Foam tape	1mm closed-cell x1m	Motor and board padding; never cover regulator surfaces
1	Epoxy	Small two-part plastic-compatible adhesive	Non-load-bearing decorative eye and panels
1	Paint and masking tape	Blue silver white black water-based	Decorative finish; keep mass low
2	PLA filament	1kg white or silver	Thin cosmetic shell; estimate includes support waste
2	PETG filament	1kg white or gray	Structural prints; validate sliced quantity

Print manifest

Solid material bound 2924.6 g, before infill savings and without supports or purchased parts. Do not use this number as measured print or assembled mass. All dimensions below are millimetres. Use the native STL orientation and the reflection instructions in the mechanical chapter.

Part	Qty	Material	X x Y x Z	Solid g each
coupon	1	PETG	90.0 x 50.0 x 5.0	25.48
body_quarter	4	PLA	130.0 x 130.0 x 140.0	43.13
body_upper_quarter	4	PLA	130.0 x 130.0 x 140.0	41.33
frame_quarter	12	PETG	130.0 x 130.0 x 5.0	34.56
splice	24	PETG	24.0 x 20.0 x 3.0	1.76
body_post	4	PETG	10.0 x 10.0 x 137.0	10.84
head_support	4	PETG	10.0 x 10.0 x 97.0	7.72
top_post	4	PETG	10.0 x 10.0 x 36.0	2.87
foot_deck	3	PETG	110.0 x 220.0 x 10.0	131.65
foot_cover	2	PLA	112.0 x 222.0 x 28.0	64.77
motor_cradle	6	PETG	44.0 x 84.5 x 49.0	15.75
leg_segment	4	PETG	50.0 x 60.0 x 125.0	62.84
shoulder_bridge	2	PETG	120.0 x 64.0 x 18.0	48.85
shoulder_cap	2	PLA	58.0 x 68.0 x 55.0	25.66
rear_bracket	1	PETG	120.0 x 146.0 x 55.0	129.89
rear_attach	1	PETG	160.0 x 42.0 x 5.0	42.07
bearing_tower	2	PETG	58.0 x 58.0 x 26.0	27.68
bearing_cap	2	PETG	34.0 x 34.0 x 3.0	2.59
race_spacer	2	PETG	11.5 x 11.5 x 12.0	0.75
spindle_sleeve	1	PETG	12.0 x 12.0 x 38.0	2.74
gear_hub	2	PETG	62.99 x 62.99 x 19.0	24.65
servo_pinion	2	PETG	32.99 x 32.99 x 6.0	4.69
servo_mount	2	PETG	36.0 x 80.0 x 26.0	24.45
servo_shim_1	2	PETG	20.0 x 40.0 x 1.0	1.02
servo_shim_2	2	PETG	20.0 x 40.0 x 2.0	2.03
servo_shim_4	2	PETG	20.0 x 40.0 x 4.0	4.06
head_deck	1	PETG	256.0 x 256.0 x 4.0	83.78
neck	1	PLA	260.0 x 260.0 x 12.0	37.12
head_plate	1	PETG	246.0 x 246.0 x 3.0	22.11
dome_spacer	4	PETG	8.0 x 8.0 x 6.6	0.35

Part	Qty	Material	X x Y x Z	Solid g each
dome	1	PLA	260.0 x 260.0 x 129.84	135.22
eye	1	PLA	32.0 x 32.0 x 20.0	8.41
detail_panel	4	PLA	72.0 x 32.0 x 2.0	4.79
battery_tray	1	PETG	124.0 x 80.0 x 15.0	57.36
utility_deck	1	PETG	170.0 x 120.0 x 3.0	70.01
deck_adapter	2	PETG	170.0 x 160.0 x 3.0	82.64
pcb_spacer	32	PETG	7.0 x 7.0 x 6.0	0.25
speaker_mount	1	PETG	88.0 x 88.0 x 12.0	22.66
switch_plate	1	PETG	70.0 x 42.0 x 3.0	10.14

Point-to-point wiring schedule

One row is one connection. Join identical net names. AWG identifies wire gauge; read the complete electrical chapter before energizing. Unused AHCT125 outputs pin8 and pin11 remain open.

Net	From	To	AWG
PACK+	B1 PP30 +	F1 7.5A input	18
FUSED+	F1 output	S1 MAIN input	18
MAIN+	S1 output	S2 RUN input	18
MAIN+	S1 output	F6 1A input	22
LOGIC_IN	F6 output	U6 MPM3610 VIN	22
RUN+	S2 output	F2 2A input	18
BUCK1_IN	F2 output	P1 UBEC IN red	22
BUCK1_IN	F2 output	CI1 220uF25V +	22
GND	B1 PP30 -	P1 UBEC IN black	18
GND	B1 PP30 -	P1 UBEC OUT black	18
GND	B1 PP30 -	CI1 -	22
RUN+	S2 output	F3 2A input	18
BUCK2_IN	F3 output	P2 UBEC IN red	22
BUCK2_IN	F3 output	CI2 220uF25V +	22
GND	B1 PP30 -	P2 UBEC IN black	18
GND	B1 PP30 -	P2 UBEC OUT black	18
GND	B1 PP30 -	CI2 -	22
RUN+	S2 output	F4 2A input	18
BUCK3_IN	F4 output	P3 UBEC IN red	22
BUCK3_IN	F4 output	CI3 220uF25V +	22
GND	B1 PP30 -	P3 UBEC IN black	18
GND	B1 PP30 -	P3 UBEC OUT black	18
GND	B1 PP30 -	CI3 -	22
RUN+	S2 output	F5 2A input	18
BUCK4_IN	F5 output	P4 UBEC IN red	22
BUCK4_IN	F5 output	CI4 220uF25V +	22
GND	B1 PP30 -	P4 UBEC IN black	18
GND	B1 PP30 -	P4 UBEC OUT black	18
GND	B1 PP30 -	CI4 -	22
5V_LEFT	P1 UBEC OUT red	D1 DRV8833 VMotor +	22

Net	From	To	AWG
5V_LEFT	P1 UBEC OUT red	CO1 470uF10V +	22
GND	B1 PP30 -	D1 GND	22
GND	B1 PP30 -	CO1 -	22
LEFT_FWD	U1 GPIO14	D1 AIN1	26
LEFT_FWD	U1 GPIO14	D1 BIN1	26
LEFT_FWD	U1 GPIO14	R_LEFT_FWD 10k pin1	26
GND	B1 PP30 -	R_LEFT_FWD pin2	26
LEFT_REV	U1 GPIO32	D1 AIN2	26
LEFT_REV	U1 GPIO32	D1 BIN2	26
LEFT_REV	U1 GPIO32	R_LEFT_REV 10k pin1	26
GND	B1 PP30 -	R_LEFT_REV pin2	26
ENABLE	U1 GPIO13	D1 SLP	26
FAULT	U1 GPIO36 / A4	D1 FLT	26
M1+	D1 AOUT1	M1 red	22
M1-	D1 AOUT2	M1 black	22
M1+	M1 red	CM1 100nF pin1	26
M1-	M1 black	CM1 pin2	26
M2+	D1 BOUT1	M2 red	22
M2-	D1 BOUT2	M2 black	22
M2+	M2 red	CM2 100nF pin1	26
M2-	M2 black	CM2 pin2	26
5V_RIGHT	P2 UBEC OUT red	D2 DRV8833 VMotor +	22
5V_RIGHT	P2 UBEC OUT red	CO2 470uF10V +	22
GND	B1 PP30 -	D2 GND	22
GND	B1 PP30 -	CO2 -	22
RIGHT_FWD	U1 GPIO15	D2 AIN1	26
RIGHT_FWD	U1 GPIO15	D2 BIN1	26
RIGHT_FWD	U1 GPIO15	R_RIGHT_FWD 10k pin1	26
GND	B1 PP30 -	R_RIGHT_FWD pin2	26
RIGHT_REV	U1 GPIO33	D2 AIN2	26
RIGHT_REV	U1 GPIO33	D2 BIN2	26
RIGHT_REV	U1 GPIO33	R_RIGHT_REV 10k pin1	26
GND	B1 PP30 -	R_RIGHT_REV pin2	26
ENABLE	U1 GPIO13	D2 SLP	26

Net	From	To	AWG
FAULT	U1 GPIO36 / A4	D2 FLT	26
M3+	D2 AOUT1	M3 red	22
M3-	D2 AOUT2	M3 black	22
M3+	M3 red	CM3 100nF pin1	26
M3-	M3 black	CM3 pin2	26
M4+	D2 BOUT1	M4 red	22
M4-	D2 BOUT2	M4 black	22
M4+	M4 red	CM4 100nF pin1	26
M4-	M4 black	CM4 pin2	26
5V_REAR	P3 UBEC OUT red	D3 DRV8833 VMotor +	22
5V_REAR	P3 UBEC OUT red	CO3 470uF10V +	22
GND	B1 PP30 -	D3 GND	22
GND	B1 PP30 -	CO3 -	22
REAR_FWD	U1 GPIO27	D3 AIN1	26
REAR_FWD	U1 GPIO27	D3 BIN1	26
REAR_FWD	U1 GPIO27	R_REAR_FWD 10k pin1	26
GND	B1 PP30 -	R_REAR_FWD pin2	26
REAR_REV	U1 GPIO12	D3 AIN2	26
REAR_REV	U1 GPIO12	D3 BIN2	26
REAR_REV	U1 GPIO12	R_REAR_REV 10k pin1	26
GND	B1 PP30 -	R_REAR_REV pin2	26
ENABLE	U1 GPIO13	D3 SLP	26
FAULT	U1 GPIO36 / A4	D3 FLT	26
M5+	D3 AOUT1	M5 red	22
M5-	D3 AOUT2	M5 black	22
M5+	M5 red	CM5 100nF pin1	26
M5-	M5 black	CM5 pin2	26
M6+	D3 BOUT1	M6 red	22
M6-	D3 BOUT2	M6 black	22
M6+	M6 red	CM6 100nF pin1	26
M6-	M6 black	CM6 pin2	26
ENABLE	U1 GPIO13	R_ENABLE 10k pin1	26
GND	B1 PP30 -	R_ENABLE pin2	26
FAULT	U1 GPIO36 / A4	R_FAULT 10k pin1	26

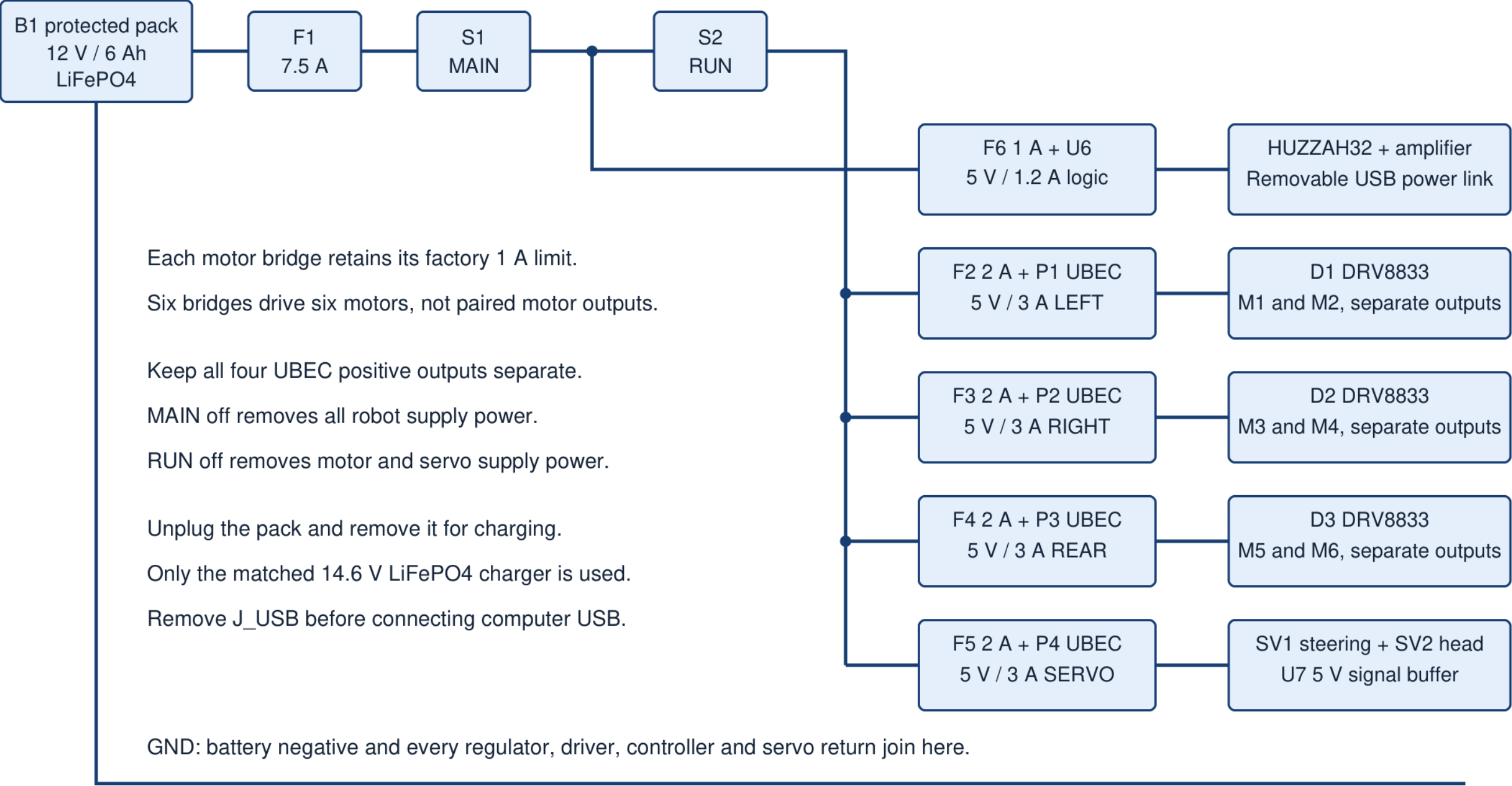
Net	From	To	AWG
3V3	U1 3V	R_FAULT pin2	26
5V_SERVO	P4 UBEC OUT red	SV1 steering red	22
5V_SERVO	P4 UBEC OUT red	SV2 head red	22
5V_SERVO	P4 UBEC OUT red	U7 AHCT125 pin14	26
5V_SERVO	P4 UBEC OUT red	CS 1000uF10V +	22
5V_SERVO	P4 UBEC OUT red	CB 100nF pin1	26
5V_SERVO	P4 UBEC OUT red	R_POWER_TOP 10k pin1	26
GND	B1 PP30 -	SV1 brown/black	22
GND	B1 PP30 -	SV2 brown/black	22
GND	B1 PP30 -	U7 pin7	26
GND	B1 PP30 -	U7 pin1 / 10E	26
GND	B1 PP30 -	U7 pin4 / 20E	26
GND	B1 PP30 -	U7 pin9 / 3A	26
GND	B1 PP30 -	U7 pin12 / 4A	26
GND	B1 PP30 -	CS -	26
GND	B1 PP30 -	CB pin2	26
GND	B1 PP30 -	R_POWER_BOTTOM 15k pin2	26
5V_SERVO	P4 UBEC OUT red	U7 pin10 / 30E	26
5V_SERVO	P4 UBEC OUT red	U7 pin13 / 40E	26
STEER	U1 GPIO25	U7 pin2	26
STEER	U1 GPIO25	R_STEER 10k pin1	26
GND	B1 PP30 -	R_STEER pin2	26
STEER_5V	U7 pin3	RS1 220ohm pin1	26
STEER_SIG	RS1 pin2	SV1 signal	26
HEAD	U1 GPIO26	U7 pin5	26
HEAD	U1 GPIO26	R_HEAD 10k pin1	26
GND	B1 PP30 -	R_HEAD pin2	26
HEAD_5V	U7 pin6	RS2 220ohm pin1	26
HEAD_SIG	RS2 pin2	SV2 signal	26
RUN_SENSE	R_POWER_TOP pin2	U1 GPIO39 / A3	26
RUN_SENSE	R_POWER_TOP pin2	R_POWER_BOTTOM pin1	26
RUN_SENSE	R_POWER_TOP pin2	CP 100nF pin1	26
GND	B1 PP30 -	CP pin2	26
MAIN+	S1 output	R_PACK_TOP 100k pin1	26

Net	From	To	AWG
PACK_ADC	R_PACK_TOP pin2	U1 GPIO34 / A2	26
PACK_ADC	R_PACK_TOP pin2	R_PACK_BOTTOM 22k pin1	26
PACK_ADC	R_PACK_TOP pin2	CA 100nF pin1	26
GND	B1 PP30 -	R_PACK_BOTTOM pin2	26
GND	B1 PP30 -	CA pin2	26
LOGIC_IN	F6 output	CI5 100uF25V +	22
GND	B1 PP30 -	CI5 -	22
GND	B1 PP30 -	U6 GND	22
5V_LOGIC	U6 5V OUT	J_USB removable link input	22
5V_LOGIC_USB	J_USB output	U1 USB pin	22
5V_LOGIC	U6 5V OUT	U8 MAX98357A VIN	22
5V_LOGIC	U6 5V OUT	CO4 470uF10V +	22
5V_LOGIC	U6 5V OUT	R_GAIN 100k pin1	22
GND	B1 PP30 -	U1 GND	22
GND	B1 PP30 -	U8 GND	22
GND	B1 PP30 -	CO4 -	22
I2S_BCLK	U1 GPIO18	U8 BCLK	26
I2S_LRC	U1 GPIO19	U8 LRC	26
I2S_DIN	U1 GPIO23	U8 DIN	26
GAIN	R_GAIN pin2	U8 GAIN	26
SPK+	U8 speaker +	SP1 +	22
SPK-	U8 speaker -	SP1 -	22

Circuit sheets

R2-24 power distribution

Architecture sheet. Detailed pin connections and capacitor polarities are on the following sheets and in wiring.csv.



Fuses are on converter inputs. Fit the specified input/output capacitors. This is not a layout drawing; use the exact wiring schedule.

00-power-overview.svg | Named nets connect across sheets. Full return wiring is in the schedule.

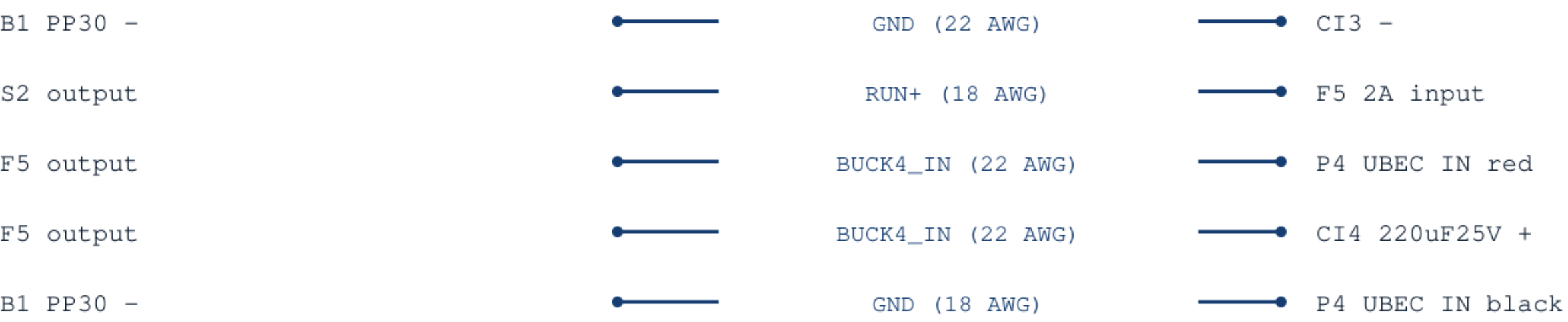
R2-24 circuit: power / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

B1 PP30 +	●	PACK+ (18 AWG)	●	F1 7.5A input
F1 output	●	FUSED+ (18 AWG)	●	S1 MAIN input
S1 output	●	MAIN+ (18 AWG)	●	S2 RUN input
S1 output	●	MAIN+ (22 AWG)	●	F6 1A input
F6 output	●	LOGIC_IN (22 AWG)	●	U6 MPM3610 VIN
S2 output	●	RUN+ (18 AWG)	●	F2 2A input
F2 output	●	BUCK1_IN (22 AWG)	●	P1 UBEC IN red
F2 output	●	BUCK1_IN (22 AWG)	●	CI1 220uF25V +
B1 PP30 -	●	GND (18 AWG)	●	P1 UBEC IN black
B1 PP30 -	●	GND (18 AWG)	●	P1 UBEC OUT black
B1 PP30 -	●	GND (22 AWG)	●	CI1 -
S2 output	●	RUN+ (18 AWG)	●	F3 2A input
F3 output	●	BUCK2_IN (22 AWG)	●	P2 UBEC IN red
F3 output	●	BUCK2_IN (22 AWG)	●	CI2 220uF25V +
B1 PP30 -	●	GND (18 AWG)	●	P2 UBEC IN black
B1 PP30 -	●	GND (18 AWG)	●	P2 UBEC OUT black
B1 PP30 -	●	GND (22 AWG)	●	CI2 -
S2 output	●	RUN+ (18 AWG)	●	F4 2A input
F4 output	●	BUCK3_IN (22 AWG)	●	P3 UBEC IN red
F4 output	●	BUCK3_IN (22 AWG)	●	CI3 220uF25V +
B1 PP30 -	●	GND (18 AWG)	●	P3 UBEC IN black
B1 PP30 -	●	GND (18 AWG)	●	P3 UBEC OUT black

R2-24 circuit: power / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.



01-power-2.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: drive / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

P1 UBEC OUT red	●————	5V_LEFT (22 AWG)	————●	D1 DRV8833 VMotor +
P1 UBEC OUT red	●————	5V_LEFT (22 AWG)	————●	CO1 470uF10V +
U1 GPIO14	●————	LEFT_FWD (26 AWG)	————●	D1 AIN1
U1 GPIO14	●————	LEFT_FWD (26 AWG)	————●	D1 BIN1
U1 GPIO14	●————	LEFT_FWD (26 AWG)	————●	R_LEFT_FWD 10k pin1
U1 GPIO32	●————	LEFT_REV (26 AWG)	————●	D1 AIN2
U1 GPIO32	●————	LEFT_REV (26 AWG)	————●	D1 BIN2
U1 GPIO32	●————	LEFT_REV (26 AWG)	————●	R_LEFT_REV 10k pin1
U1 GPIO13	●————	ENABLE (26 AWG)	————●	D1 SLP
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	D1 FLT
D1 AOUT1	●————	M1+ (22 AWG)	————●	M1 red
D1 AOUT2	●————	M1- (22 AWG)	————●	M1 black
M1 red	●————	M1+ (26 AWG)	————●	CM1 100nF pin1
M1 black	●————	M1- (26 AWG)	————●	CM1 pin2
D1 BOUT1	●————	M2+ (22 AWG)	————●	M2 red
D1 BOUT2	●————	M2- (22 AWG)	————●	M2 black
M2 red	●————	M2+ (26 AWG)	————●	CM2 100nF pin1
M2 black	●————	M2- (26 AWG)	————●	CM2 pin2
P2 UBEC OUT red	●————	5V_RIGHT (22 AWG)	————●	D2 DRV8833 VMotor +
P2 UBEC OUT red	●————	5V_RIGHT (22 AWG)	————●	CO2 470uF10V +
U1 GPIO15	●————	RIGHT_FWD (26 AWG)	————●	D2 AIN1
U1 GPIO15	●————	RIGHT_FWD (26 AWG)	————●	D2 BIN1

R2-24 circuit: drive / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

U1 GPIO15		RIGHT_FWD (26 AWG)	R_RIGHT_FWD 10k pin1
U1 GPIO33		RIGHT_REV (26 AWG)	D2 AIN2
U1 GPIO33		RIGHT_REV (26 AWG)	D2 BIN2
U1 GPIO33		RIGHT_REV (26 AWG)	R_RIGHT_REV 10k pin1
U1 GPIO13		ENABLE (26 AWG)	D2 SLP
U1 GPIO36 / A4		FAULT (26 AWG)	D2 FLT
D2 AOUT1		M3+ (22 AWG)	M3 red
D2 AOUT2		M3- (22 AWG)	M3 black
M3 red		M3+ (26 AWG)	CM3 100nF pin1
M3 black		M3- (26 AWG)	CM3 pin2
D2 BOUT1		M4+ (22 AWG)	M4 red
D2 BOUT2		M4- (22 AWG)	M4 black
M4 red		M4+ (26 AWG)	CM4 100nF pin1
M4 black		M4- (26 AWG)	CM4 pin2
P3 UBEC OUT red		5V_REAR (22 AWG)	D3 DRV8833 VMotor +
P3 UBEC OUT red		5V_REAR (22 AWG)	CO3 470uF10V +
U1 GPIO27		REAR_FWD (26 AWG)	D3 AIN1
U1 GPIO27		REAR_FWD (26 AWG)	D3 BIN1
U1 GPIO27		REAR_FWD (26 AWG)	R_REAR_FWD 10k pin1
U1 GPIO12		REAR_REV (26 AWG)	D3 AIN2
U1 GPIO12		REAR_REV (26 AWG)	D3 BIN2
U1 GPIO12		REAR_REV (26 AWG)	R_REAR_REV 10k pin1

R2-24 circuit: drive / 3

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

U1 GPIO13	●————	ENABLE (26 AWG)	————●	D3 SLP
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	D3 FLT
D3 AOUT1	●————	M5+ (22 AWG)	————●	M5 red
D3 AOUT2	●————	M5- (22 AWG)	————●	M5 black
M5 red	●————	M5+ (26 AWG)	————●	CM5 100nF pin1
M5 black	●————	M5- (26 AWG)	————●	CM5 pin2
D3 BOUT1	●————	M6+ (22 AWG)	————●	M6 red
D3 BOUT2	●————	M6- (22 AWG)	————●	M6 black
M6 red	●————	M6+ (26 AWG)	————●	CM6 100nF pin1
M6 black	●————	M6- (26 AWG)	————●	CM6 pin2
U1 GPIO13	●————	ENABLE (26 AWG)	————●	R_ENABLE 10k pin1
U1 GPIO36 / A4	●————	FAULT (26 AWG)	————●	R_FAULT 10k pin1

02-drive-3.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: servo-and-sense / 1

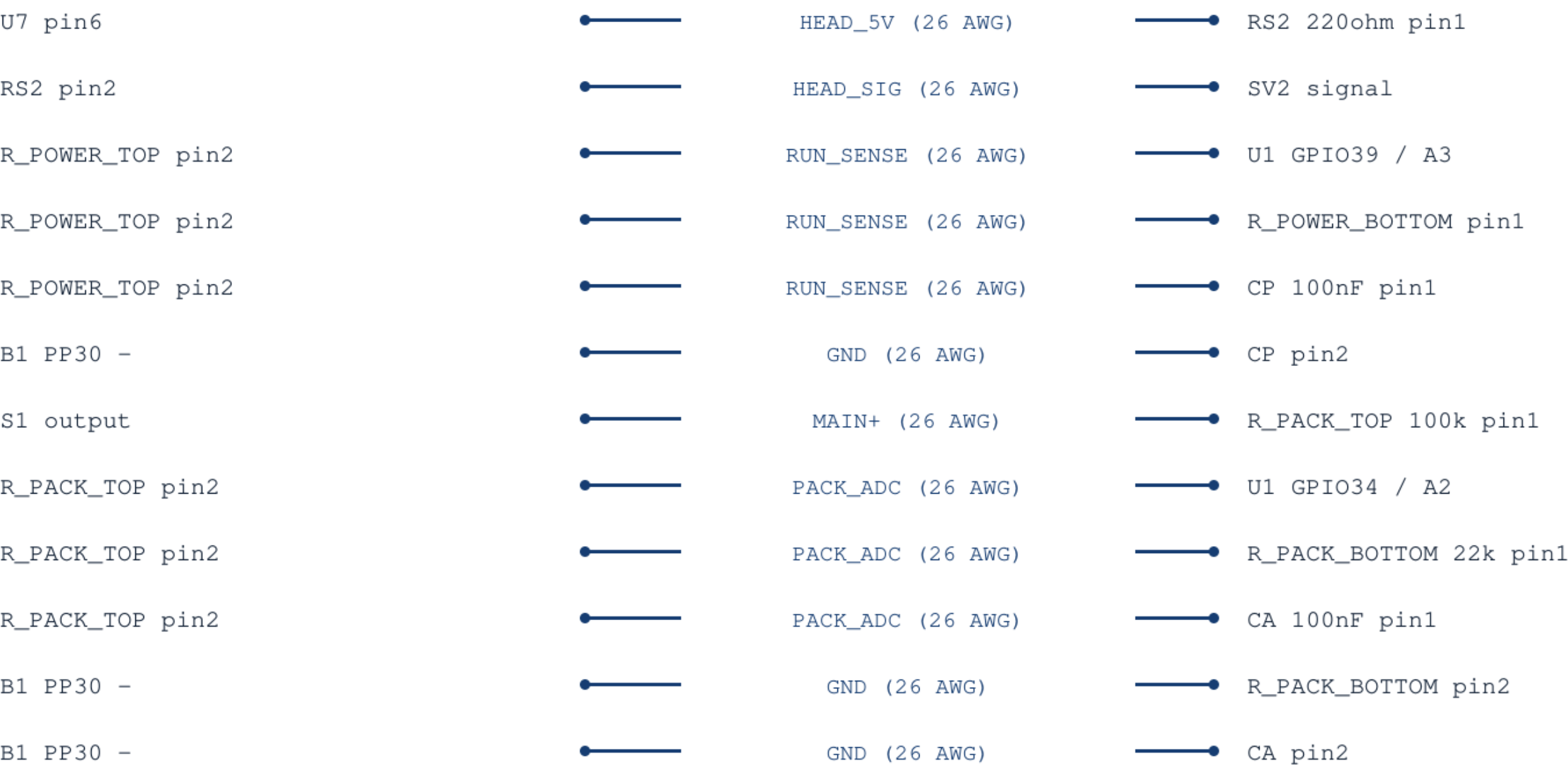
Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

P4 UBEC OUT red	●—————	5V_SERVO (22 AWG)	—————●	SV1 steering red
P4 UBEC OUT red	●—————	5V_SERVO (22 AWG)	—————●	SV2 head red
P4 UBEC OUT red	●—————	5V_SERVO (26 AWG)	—————●	U7 AHCT125 pin14
P4 UBEC OUT red	●—————	5V_SERVO (22 AWG)	—————●	CS 1000uF10V +
P4 UBEC OUT red	●—————	5V_SERVO (26 AWG)	—————●	CB 100nF pin1
P4 UBEC OUT red	●—————	5V_SERVO (26 AWG)	—————●	R_POWER_TOP 10k pin1
B1 PP30 -	●—————	GND (22 AWG)	—————●	SV1 brown/black
B1 PP30 -	●—————	GND (22 AWG)	—————●	SV2 brown/black
B1 PP30 -	●—————	GND (26 AWG)	—————●	U7 pin7
B1 PP30 -	●—————	GND (26 AWG)	—————●	U7 pin1 / 10E
B1 PP30 -	●—————	GND (26 AWG)	—————●	U7 pin4 / 20E
B1 PP30 -	●—————	GND (26 AWG)	—————●	U7 pin9 / 3A
B1 PP30 -	●—————	GND (26 AWG)	—————●	U7 pin12 / 4A
B1 PP30 -	●—————	GND (26 AWG)	—————●	R_POWER_BOTTOM 15k pin2
P4 UBEC OUT red	●—————	5V_SERVO (26 AWG)	—————●	U7 pin10 / 30E
P4 UBEC OUT red	●—————	5V_SERVO (26 AWG)	—————●	U7 pin13 / 40E
U1 GPIO25	●—————	STEER (26 AWG)	—————●	U7 pin2
U1 GPIO25	●—————	STEER (26 AWG)	—————●	R_STEER 10k pin1
U7 pin3	●—————	STEER_5V (26 AWG)	—————●	RS1 220ohm pin1
RS1 pin2	●—————	STEER_SIG (26 AWG)	—————●	SV1 signal
U1 GPIO26	●—————	HEAD (26 AWG)	—————●	U7 pin5
U1 GPIO26	●—————	HEAD (26 AWG)	—————●	R_HEAD 10k pin1

03-servo-and-sense-1.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: servo-and-sense / 2

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.



03-servo-and-sense-2.svg | Named nets connect across sheets. Full return wiring is in the schedule.

R2-24 circuit: logic-and-audio / 1

Each horizontal line is one connection. Identical named nets join across sheets. See wiring.csv for all grounds.

F6 output	●————	LOGIC_IN (22 AWG)	————●	U6 MPM3610 VIN
F6 output	●————	LOGIC_IN (22 AWG)	————●	CI5 100uF25V +
B1 PP30 -	●————	GND (22 AWG)	————●	CI5 -
B1 PP30 -	●————	GND (22 AWG)	————●	U6 GND
U6 5V OUT	●————	5V_LOGIC (22 AWG)	————●	J_USB removable link input
J_USB output	●————	5V_LOGIC_USB (22 AWG)	————●	U1 USB pin
U6 5V OUT	●————	5V_LOGIC (22 AWG)	————●	U8 MAX98357A VIN
U6 5V OUT	●————	5V_LOGIC (22 AWG)	————●	CO4 470uF10V +
U6 5V OUT	●————	5V_LOGIC (22 AWG)	————●	R_GAIN 100k pin1
B1 PP30 -	●————	GND (22 AWG)	————●	U1 GND
B1 PP30 -	●————	GND (22 AWG)	————●	U8 GND
B1 PP30 -	●————	GND (22 AWG)	————●	CO4 -
U1 GPIO18	●————	I2S_BCLK (26 AWG)	————●	U8 BCLK
U1 GPIO19	●————	I2S_LRC (26 AWG)	————●	U8 LRC
U1 GPIO23	●————	I2S_DIN (26 AWG)	————●	U8 DIN
R_GAIN pin2	●————	GAIN (26 AWG)	————●	U8 GAIN
U8 speaker +	●————	SPK+ (22 AWG)	————●	SP1 +
U8 speaker -	●————	SPK- (22 AWG)	————●	SP1 -

04-logic-and-audio-1.svg | Named nets connect across sheets. Full return wiring is in the schedule.